

MUHAMMAD DANYAL ('SAGE') KHAN

Marie Skłodowska-Curie PhD Scholar · AI, Cyber Security and Blockchain Scientist · Systems Thinker · Published Technical and Fiction Author · Rhetorician · Polymath · Polyglot · FOSS Advocate · CloudNative & DataOps (CD Foundation) · DevOps & DevRel · GRC Professional · Cyber Warrior · Ex-Naval Officer · Educator · Audio Engineer · Musician · Hypnotherapist · Open-Source Technologist · Lead Auditor · Entrepreneur · Meditation & Life Coach · Speed Learner · Silva UM Instructor



📍 Granada, Spain

🌐 github.com/sage-khan

📅 [google-calendar/JpzowQkFj39zH9zu7](https://calendar.google.com/calendar/u/0?cid=JpzowQkFj39zH9zu7)

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Professional Summary

*A Dynamic, cross-domain system thinker and versatile professional with cross-domain expertise in **Cybersecurity, Artificial Intelligence, DevOps, and Open-Source Systems**. Marie Skłodowska-Curie PhD Scholar specializing in the convergence of Artificial Intelligence, Cybersecurity, and Decentralized Infrastructure. Bridged the gap between mission-critical physical assets (autonomous wind farms, naval platforms) and advanced digital logic (agentic LLM frameworks, blockchain ontologies). Leveraging a dual background as a former Naval Officer and a Senior AI Developer, architected "trust-aware" systems that prioritize security, explainability, and scalability. A visionary leader experienced in guiding cross-cultural, cross-domain teams across **military, academic, and corporate ecosystems**. As a Project Lead for the Linux Foundation's CD Foundation, defined the frameworks that shape the global DataOps landscape. My work—ranging from published technical authorship to the development of polyglot-persistence agentic databases (Project Veritas)—is driven by a commitment to technological sovereignty, open-source advocacy, and the human-centric leadership required to guide cross-domain teams through complex digital transformations. Demonstrated a proven ability to deliver **secure, scalable, and high-impact AI solutions** at the intersection of technology, leadership, and education. Published author of the book "Open for Business," exploring the integration of open-source technologies into corporate IT infrastructures. Regularly writes content on cybersecurity, AI, and self-help, showcasing a commitment to bridging technical expertise with practical insights. Driven by a passion for secure and scalable digital ecosystems, contribute to the intersection of technology and sustainability through innovative research and thought leadership. Featured as an **AI & Cybersecurity Expert** on Cloud Native FM Podcast and **Frontline with Tazmeen Javaid (News 99)**.*

Skills

Spoken Languages: English (C1 R/W/S), Urdu (C1 R/W/S), Arabic (A2 R/W/S), Russian (A1 R/W/S), Spanish (A1 R/W/S), Punjabi (S), Pushto (S)

Programming Languages: Python, R, C++, Bash, Java, SQL, HTML/CSS, JavaScript, LaTeX, Markdown. Skilled in LLM prompting, fine-tuning, and integration across models such as GPT, LLaMA, Mistral, and LLaVA.

AI & Machine Learning Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, LangChain, LlamaIndex, OpenLLM, LM Studio, Ollama, Unsloth, LlamaFactory. Hands-on experience with model deployment, RAG pipelines, and fine-tuning.

Data Science & Analytics: Data Analysis, NLP, Generative AI, RAG, ETL, Data Warehousing, Business Intelligence, Tesseract OCR, OCRmyPDF, Power BI, Jupyter, Anaconda.

ETL & Workflow Automation Tools: Airflow, Airbyte, PySpark, Knime, Unstructured.io, n8n.io, Crew AI.

Cloud & Infrastructure Platforms: AWS, Azure, Google Cloud Platform (GCP), Oracle Cloud, Digital Ocean, RunPod, Proxmox, OpenStack, VMware. Proficient in containerized and virtualized environments.

DevOps & MLOps: Docker, Kubernetes, Terraform, Git, CI/CD (GitHub Actions, Jenkins), AWS SageMaker, MLflow, Kaggle. Experience in automated deployment pipelines and model lifecycle management.

Cybersecurity & GRC Frameworks: NIST CSF, ISO 27001, SOC 2, PCI DSS, HIPAA, Eramba, JupiterOne. Skilled in InfoSec audits, Governance, Risk Management, Compliance, and Incident Response.

Technologies & Tools: Linux (Ubuntu, Kali, Fedora, CentOS), WordPress, ERPNext, RPA Framework, PyAutoGUI, MeshCentral, F5/E2 TTS, Pyannote Diarizer, Speechbrain, Kaldi, Vosk, DeepSpeech.

Database Technologies: MySQL, PostgreSQL, MS SQL Server, MongoDB, DynamoDB, CosmosDB, Vector Databases (Chroma, Qdrant, Weaviate, Pinecone), Blob Storage.

Developer Tools: VS Code, Windsurf, Cursor AI, Jupyter, Anaconda, LaTeX, Markdown, REST API development.
Administration & Productivity Tools: Microsoft Office Suite, Excel (advanced formulas and dashboards), Trello, Jira (Agile & Scrum), Confluence, Notion.
Audio Engineering & Music Production: Digital Audio Workstations (FL Studio, Ardour), Audio Editing (Audacity), Mixing & Mastering, Sound Design, Music Composition, Piano, Guitar, Vocals.
Soft Skills: Leadership, Strategic Planning, Mentorship, Public Speaking, Technical Documentation, Research Writing, Problem solving, Team Building, Workshop Facilitation.
Video & Content Creation: OBS Studio, OpenShot, Canva, StreamYard. Experienced in the creation of educational content, live training, and online broadcasting.

Job Experience

IPM Lab Universidad de Granada **1 Sept 2025 – Present**
MSCA Personal Docente e Investigador *Onsite*

- Research on topic **Secure IT framework for autonomous wind farm operation and maintenance based on blockchain, smart contracts and AI** under Dr. Manuel Chiachio Ruano and Dr. Juan Chiachio Ruano, along with Dr. Juan Fernandez (QUANTIA ENGINEERING & CONSULTANCY, S.L.) for development of a secure, trustworthy, explainable, and interactive structural health monitoring (SHM) platform for wind farms combining AI algorithms, decentralized knowledge graphs, and blockchain.
- Collaborated with Industry Partners, i.e., NADARA and Quantia, to develop an efficient multi-modal Database with blockchain for wind farm and solar farm data.
- Development of MVP of Project Veritas, the blockchain-powered ontology aware agentic Polyglot Persistence Database.

Founder and Lead Architect **11 Feb 2026 – Present**
Veritas AI DB *Remote, Granada, Spain*

- Architected ontology-driven polyglot data platform integrating 7 heterogeneous databases, relational, graph, vector, search, time-series, object storage, blockchain.
- Designed trust-scored routing layer with uncertainty-aware confidence bounds for data governance.
- Implemented GraphRAG architecture for industrial SCADA and document intelligence workloads.
- Built 13-step recursive industrial document processing pipeline, multi-engine OCR, layout analysis, semantic chunking, embedding generation.
- Developed agentic data quality framework with distributed tracing via OpenTelemetry for non-deterministic workflows.
- Implemented global UUID-based provenance system spanning polyglot stores.
- Open-sourced platform under the GitHub organization. Actively developing as thesis-backed research and product.

Project Lead – DataOps Initiative **Mar 2025 – Present**
Continuous Delivery Foundation · Freelance *Remote, United States*

- Led the Continuous Delivery Foundation’s **DataOps Landscape Initiative**, defining its vision, roadmap, and governance to strengthen the open-source DataOps ecosystem.
- Developed a structured framework of **categories, tools, and technologies** within the DataOps landscape to enhance visibility and accessibility for contributors and organizations.
- Scoped and curated **example projects and reference implementations** that demonstrate practical DataOps use cases and best practices.
- Designed and implemented **evaluation criteria and inclusion standards** to ensure project quality, maintainability, and community relevance.
- Engaged with the global DataOps and CI/CD community to **encourage project submissions** via GitHub and increase participation in the foundation’s ecosystem.
- Established and maintained high-quality guidelines for project inclusion, emphasizing **technical maturity, documentation quality, and sustainability**.
- Authored and published **technical blog posts and thought leadership content** to promote awareness and adoption of DataOps principles.
- Organized and facilitated **community workshops and webinars** to educate professionals on the DataOps landscape, frameworks, and industry impact.
- Reviewed and edited technical content to ensure **clarity, consistency, and accuracy** across foundation publications and materials.

Syllaby AI **Mar 2025 – Jul 2025**
Senior AI Developer *Remote*

- Led the design and development of **scalable image generation pipelines** using Flux and SDXL, integrating cloud APIs (Replicate) and on-prem deployments to optimize inference cost and latency.

- Formed and managed a **cross-functional engineering team**, building Dockerized microservices and RESTful APIs to support backend integrations and accelerate product delivery.
- Acted as a **technical product owner**, defining feature requirements, managing agile sprints, and delivering production-grade AI systems informed by real user feedback.
- Developed **advanced ComfyUI workflows** for consistent character and scene generation, modularizing nodes for reusability by non-technical stakeholders.
- Implemented **ControlNet with OpenPose and depth maps** to enforce pose and scene consistency, improving image coherence and creative control.
- Fine-tuned Flux diffusion models using **Low-Rank Adaptation (LoRA)**, reducing compute cost while improving model fidelity and domain specialization.
- Deployed **ML inference systems** using Django, Docker, and Kubernetes on DigitalOcean, integrating CI/CD pipelines via GitHub Actions for reliable production scaling.
- Enhanced **voice generation and speech pipelines** by integrating ElevenLabs, Hugging Face, and Replicate models, improving accuracy and latency for real-time agents.
- Delivered full-stack Django applications with versioned REST APIs, JSON Web Token Authentication, caching, and monitoring to support both internal and external clients.
- Created **text-to-video generation systems** by combining D-ID and Dia TTS, enabling lifelike talking head experiences with synchronized speech and facial animation.
- Built an **agentic brainstorming platform** using **n8n.io**, integrating LLM agents, vector databases, and API orchestration to automate internal ideation workflows.

TwinCiti AI

Dec 2024 – Jan 2025

Machine Learning Engineer (Contract)

Remote

- Developed and integrated **Robotic Process Automation (RPA)** workflows using AutoGUI, Selenium, Playwright, and LLMs via OpenRouter to streamline business operations.
- Implemented and deployed **ASR, STT, TTS, and voice cloning systems** leveraging F5, E2, and cutting-edge AI models for conversational assistants.
- Collaborated with globally distributed teams in an **Agile Scrum environment** using Jira, ensuring timely and high-quality AI solution delivery.
- Integrated **Large Language Models (LLMs)** through OpenRouter and ChatGPT APIs to enhance automation and reasoning capabilities across pipelines.
- Developed **speaker diarization systems** using Pyannote and Whisper-based transcript analysis to accurately separate and label speakers in multi-speaker audio.
- Delivered end-to-end **AI model training, optimization, and deployment**, ensuring scalable, maintainable, and production-ready solutions.

HIQ Solutions LLC

8 Jul 2024 – 25 Dec 2024

Machine Learning Consultant

Remote (Part-Time)

- Pre-Sales Consultation - Acted as a vital link between US Clients and the global development team
- Participating in introductory and technical meetings
- Engaging with clients to understand their requirements and conveying these to the development team
- Ensuring clear and effective communication between all parties
- Effective training and deployment of AI solutions as per client requirement

Citizen Police Liaison Committee (CPLC)

Oct 2021 – Dec 2022

AI-ASR Engineer (Volunteer Member)

Part-Time

- Engineered and trained a custom **Speech-to-Text (STT)** system from scratch using a Hybrid HMM-DNN architecture on a proprietary dataset for CPLC's voice analysis initiatives.
- Automated the end-to-end **data preprocessing and model training pipeline** using Linux services, cron jobs, and custom Bash scripts for continuous integration and maintenance.
- Utilized **Kaldi, DeepSpeech, and Vosk frameworks** to optimize speech recognition accuracy and system scalability for operational deployment.

Pakistan Navy

Jan 2012 – May 2025

Lieutenant Commander

Pakistan

- Led the **development and containerized deployment** of Analytical War Gaming, GIS, and War Simulation software using open-source technologies (Flask, SQL) on Linux servers.
- Integrated **AI and Large Language Models (LLMs)** via Ollama to automate internal processes and improve operational decision-making efficiency.

- Authored and implemented **Cyber Warfare doctrines**, covering Electronic Warfare, Intelligence Operations, and Information Operations strategy and execution.
- Conducted **information security audits, governance, and regulatory compliance assessments** across multiple naval units.
- Established and led **Incident Response and Cyber Emergency Response Teams (CERTs)**, providing Cyber Security readiness training to operational units.
- Deployed and customized **open-source software solutions** to optimize administrative workflows and automate mission-specific processes.
- Served in multiple technical and command roles, including **Signal Communication Officer, Electronic Warfare Officer, Information Warfare Officer, Principal Underwater Warfare Officer, and Executive Officer**, managing multi-disciplinary naval operations.
- Directed **maritime law enforcement and security operations**, including port and fishing control, anti-human trafficking, counter-narcotics, and anti-smuggling missions.
- Delivered **training programs for junior officers** at naval training establishments and on deployed vessels, mentoring personnel in operations, leadership, and technical skills.
- Served aboard multiple platforms including **Type-21 Destroyers, F22P Frigates, and Corvettes**, contributing to operational readiness and mission execution.
- Completed advanced naval and combat training in **SEAL operations, close-quarter combat, surface and underwater warfare, helicopter operations, anti-piracy, naval leadership, and administration**.

Independent Consultation & Freelancing

AI Algorithmic Auditing in Action | *EC-Council*

Mar 2026 – Present

- Contracted as **Author and Instructor** to design and develop a professional training course in **AI Algorithmic Auditing** within the domain of information security.
- Developing end-to-end course content including **video lectures (3–4 hours)**, structured modules, and comprehensive training materials.
- Authored full curriculum covering **Machine Learning fundamentals, Algorithmic Auditing, AI Ethics, Legal frameworks, Bias and Fairness, and Explainability (XAI)**.
- Created supporting materials including **course handbook (eBook), section-wise quizzes, final assessment (25 questions), metadata documentation, and instructional “how-to” videos**.
- Designed content aligned with **industry compliance, security standards, and real-world AI governance practices**.
- Engaged in iterative review cycles with EC-Council, incorporating **editorial and technical feedback** to meet publication standards.
- Contributed to a commercially distributed product with **royalty-based compensation model (20% of net receipts)**.

Senior AI Consultant | *Nova Tech (Built Pulse)*

Nov 2025 – Feb 2026

- Architecting the **5-Bot AI Implementation Project**, specifically developing the **Knowledge Bot** using a **GraphRAG** approach (integrating **Knowledge Graphs** with **Vector Databases**) to enable high-fidelity retrieval of complex organizational data. Deployed on <https://bots.builtpulse.com>
- Developed and enforced enterprise-grade **password management policies** aligned with **ISO 27001** standards, implementing Passbolt for secure credential orchestration and access control.
- Established robust **GitOps** workflows and automated **CI/CD pipelines** to manage containerized microservices, ensuring seamless deployment and environment parity across **AWS and Azure** clouds.
- Automated organizational **employee attendance tracking** by deploying and integrating **Kimai and TimeTrex**, streamlining HR operations and resource reporting via custom API integrations. Final deployment was on horillaHRMS. Led the deployment of customized CRM
- Designing a multi-agent orchestration framework utilizing **LangChain, CrewAI, and n8n.io**, built on a backend of **Python 3.11, FastAPI, RabbitMQ, and Redis** for scalable asynchronous processing.
- Drafting end-to-end **AI/ML product development SOPs** and security protocols for IP protection, while facilitating weekly technical workshops for core engineering teams on agentic architectures.

Deployment of Open Source SOC Using Wazuh | *Virtuous BPO*

Aug 2024 – In Progress

- Implemented an open-source **Security Operations Center (SOC)** using **Wazuh**.
- Customized detection rules and dashboards based on organizational security and compliance requirements.

Process Automation Using RPA Framework | *Virtuous BPO*

Jun 2024 – Aug 2024

- Designed and deployed **custom RPA bots** using Python to automate accounting and BPO workflows in **QuickBooks**.
- Streamlined repetitive business processes to enhance efficiency and reduce manual workloads.

Large Language Model Training Optimization | *41 Labs LLC, USA*

Mar 2024

- Provided consultation on optimization and fine-tuning of **GPT-3.5** for medical domain use cases.
- Employed **OpenAI API**, Python, Bash, JSON, and SQL for training, evaluation, and deployment workflows.

Open Source Offline LLM Deployment | *Virtuous BPO*

Jan – May 2024

- Consulted on deployment of an open-source **Large Language Model (LLM)** customized for BPO operations.
- Implemented **PrivateGPT RAG solutions** and deployed **Ollama LLM servers** on remote Linux systems.
- Configured **NVIDIA drivers, CUDA, NCCL, and cuDNN** for optimized GPU performance.
- Developed automated workflows using Python, including an offline PDF-to-Excel conversion tool integrated with PrivateGPT.
- Built **RAG pipelines** using **LangChain, LlamaIndex, Ollama**, and embeddings such as **BERT and MiniLM** with VectorDBs (Chroma, Qdrant, Weaviate, Pinecone).
- Managed model operations (push, pull, edit) for open-source models like **LLaMA and Mistral**.
- Deployed **GitLab, Jenkins, and n8n.io** for CI/CD and automation workflows.
- Developed **Crew AI-based bots** for task automation and process optimization.
- Utilized Python, Bash, JSON, SQL, MongoDB, and vector databases for end-to-end development.

Open Source ERP and HRMIS Deployment | *Virtuous BPO*

Dec 2023 – Jan 2024

- Deployed **ERPNext HR and Training modules** alongside **OrangeHRMS** for HR and operations automation.
- Configured deployments across **Ubuntu, Kali, CentOS, and Fedora** systems.
- Utilized Python, Bash, JSON, SQL, and MongoDB for integration, customization, and backend scripting.

Open Source Deployment for Efficient Office Administration | *Virtuous BPO*

Nov – Dec 2023

- Deployed open-source platforms to enhance administrative efficiency using Python, Bash, JSON, SQL, and MongoDB.
- Implemented **MeshCentral** for remote PC management and **Passbolt/TeamPass** for secure password handling.
- Configured systems on **Ubuntu, Kali, CentOS, and Fedora** environments.
- Developed **governance, regulatory, and compliance policies** aligned with **ISO 27001** and **NIST CSF**, including ACLs, cyber-physical security, and information management standards.

Open Source Network Deployment | *Virtuous BPO*

Jun – Nov 2023

- Developed and implemented **secure network infrastructure** using Python, Bash, JSON, SQL, and MongoDB.
- Configured and deployed **VPNs, SSH, and secure connections** to ensure system integrity.
- Set up **testing servers, virtual machines, and LxCs** using **Proxmox** for localized cloud environments.
- Configured multi-distro environments including **Ubuntu, Kali, CentOS, and Fedora**.

Technical Content Writing | *Disruptivera, Medium*

2020–2022

- Authored technical blogs and articles on **AI, Data Science, and emerging technologies**.
- Conducted **topic ideation, keyword analysis, proofreading, and editing** for clarity and engagement.
- Developed **technical documentation and tutorials** to explain complex AI concepts to general audiences.

Content Writing – US Concept Pages | *Upwork*

2010

- Authored SEO-optimized content for U.S.-based concept pages and marketing sites.
- Performed **keyword research** and implemented **search engine optimization (SEO)** strategies to increase visibility.
- Delivered high-quality written material aligned with client objectives and brand tone.

Freelance Audio Engineer and Music Composer | *Upwork*

2010

- Composed and produced original music tracks for diverse projects.
- Developed complete audio tracks using **FL Studio** and **Audacity**, managing the full production lifecycle.
- Performed professional **mixing and mastering** to ensure optimal sound quality and client satisfaction.

Teaching & Trainer Experience

The Profs UK Spires

Private Coach (Part-Time)

Jan 2026 – Present

Remote, United Kingdom

- One-on-one teaching in AI, Cyber Security, programming, Mathematics, Statistics, from A Levels to Grad students.

Iqra University

Lecturer (Part-Time), Project Supervisor

Aug 2024 – Present

Karachi, Pakistan

- Instructed undergraduate and postgraduate courses on **Generative AI, Critical Infrastructure Protection, Incident Response, Advanced Operating Systems, Digital Forensics, Ethical Hacking, and Information Security Governance**.
- Developed comprehensive **lab manuals, course outlines, and teaching materials** aligned with academic and industry standards.
- Delivered professional training sessions at the **Professional Development Center (PDC)** focusing on applied AI and cybersecurity skills.
- Supervised **student research and capstone projects**, guiding project planning, execution, and defense preparation.

Bahria University Karachi

Lecturer (Part-Time), Project Supervisor

Sep 2024 – Oct 2024

Karachi, Pakistan

- Taught undergraduate courses in **Management Information Systems, Data Analytics, Business Analytics, Machine Learning, and Generative AI**.
- Developed and refined **curricula and instructional content** for Business Analytics and Data Science courses.
- Supervised student projects, providing academic mentorship and technical guidance in data-driven research.

National University of Sciences and Technology (NUST)

Instructor, Junior Officer Training Officer, Head of Postgraduate Program

Sep 2020 – Dec 2022

Karachi, Pakistan

- Appointed by the **Pakistan Navy** to train newly commissioned naval officers in academic and technical disciplines.
- Designed and established **Cybersecurity laboratories** for both foundational and advanced training programs.
- Contributed to the development of **Cyber Warfare and AI doctrines** for integration into national defense education.
- Conducted **information security audits** and implemented governance, regulatory, and compliance measures in alignment with **ISO 27001, ISO 27002, and NIST CSF** frameworks.
- Performed **physical security assessments and policy enforcement** at the institutional management level.
- Delivered professional development courses in **Artificial Intelligence and Cybersecurity** at NUST's Professional Development Center.
- Taught postgraduate courses in **Digital Forensics, Ethical Hacking, and Information Security Governance**, combining theory with hands-on practice.

Education

Marie-Curie PhD Scholar

IPM Lab, Universidad de Granada

July 2025 – Present

Granada, Spain

Marie Skłodowska-Curie Actions Personal Docente e Investigador (PhD Scholar)

Research Work

- | | | |
|---------------------------|-------------------|-------------------------|
| • Artificial Intelligence | • Blockchain | • Internet of Things |
| • Sustainable Development | • Cyber Security | • Network Security |
| • Renewable Energy | • Knowledge Graph | • Structural Monitoring |

Master of Science in Cyber Security

National University of Science & Technology - Pak Navy Engineering College

Sep. 2020 – Oct. 2022

Karachi, Pakistan

CGPA 4.0 / 4.0 WITH GOLD MEDAL

Relevant Coursework

- | | | | |
|-----------------------|---------------------|---------------------------|------------------------|
| • Info Sec Management | • Digital Forensics | • Artificial Intelligence | • Applied Cryptography |
| • SDLC | • Ethical Hacking | • Internet of Things | • Network Security |

Bachelor of Science in Management Information Systems

National University of Science & Technology - Pak Navy Engineering College

Jan. 2012 – Dec. 2018

Karachi, Pakistan

CGPA 3.72 / 4.0

Relevant Coursework

- Programming
- Database
- Entrepreneurship
- Human-Computer Interaction
- Web Development
- Data Warehousing
- Organizational Behavior
- Computer Architecture
- Networking
- SDLC
- Operations Research

GCE A Levels

Bahria College Karachi NORE-I

1A* 3A 1B 1C

Aug. 2010 – Aug. 2012

Karachi, Pakistan

Subjects & Results

- Accounting (A* Grade)
- Physics (A Grade)
- Thinking Skills (B Grade)
- Mathematics (A Grade)
- Chemistry (A Grade)
- Further Maths (C Grade)

GCE O Levels

Bahria College Karachi NORE-I

10As

1996 – Aug. 2010

Karachi, Pakistan

Subjects & Results

- English (A Grade)
- Add. Math (A Grade)
- Biology (A Grade)
- Islamiat (A Grade)
- Urdu (A Grade)
- Physics (A Grade)
- Computer Science (A Grade)
- History & Geography of Pakistan (A Grade)
- Mathematics (A Grade)
- Chemistry (A Grade)

Professional Certifications

Audio Engineering & Music Production | *Audio Institute of America*

Feb 2016

- Completed professional certification in **Audio and Sound Engineering**, covering recording, mixing, and mastering techniques.
- Trained in the operation and maintenance of **studio recording equipment** and production workflows.
- Developed expertise in **music production and arrangement** for digital and analog environments.
- Gained hands-on experience with **Digital Audio Workstations (DAWs)** and post-production processes.

ISO/IEC 27001:2013 Lead Auditor | *IRCA, RICI*

Apr 2024

- Successfully completed **ISO/IEC 27001:2013 Lead Auditor Training**, accredited by **IRCA and RICI (Lahore)**.
- Credential ID: **CQI-ULN-554424 (Certificate #08983)**.
- Specialized in **Information Security Auditing, Governance, Risk Management, and Compliance (GRC)** in alignment with international standards.

Registered Information Technology Freelancer | *Pakistan Software Export Board, Ministry of IT (MoIT)*

May 2025

- Officially registered as an **IT Freelancer** with the **Ministry of IT & Telecom, Government of Pakistan**.
- Registration ID: **FL21/PSEB/2025/13053**.

Professional / Specialized Courses

IntelliWind MSCA-DN - Training School 1 | *Technical University of Denmark (DTU)*

Sep 15 – Sep 24, 2025

- Completed intensive training on the wind energy sector, focusing on data science, AI, and research applications for autonomous wind farms.
- Core Technical Topics: Wind Energy Fundamentals, Data Science essentials, Machine Learning and Deep Learning for wind, and hands-on analysis of SCADA data.
- Studied advanced concepts including AI & ML impact on operations, Prompt Engineering with LLMs, and the autonomous systems concept.
- Transferrable Skills: Attended workshops on writing scientific papers, Open Science tools, researcher communication, and PhD project management.
- Confirmed attendance and full participation for the event held at DTU Wind and Energy Systems, Roskilde, Denmark.
- ECTS Credits: Training covered 8 full days between September 15th and September 24th, 2025).

Incident Response Course | *PDC NUST-PNEC*

Apr 2021

- Completed hands-on training in **log extraction, management, and forensic imaging**.
- Studied **legal and procedural aspects** of incident response and digital forensics.
- Learned structured **reporting procedures, command, and control frameworks** for incident management.

Linux for System Administration | *PDC NUST-PNEC*

Dec 2021 – Feb 2022

- Completed practical training across multiple **Linux distributions** including Ubuntu, Kali, CentOS, and Fedora.

- Configured desktop environments such as GNOME, KDE, and XFCE, with focus on usability and performance. - Gained proficiency in package and key management (apt, dnf, snap). - Implemented automation using cron jobs, and worked with Docker and Kubernetes for containerized deployments. - Utilized curl, wget, ssh, sshfs, and scp for secure network operations. - Gained hands-on experience with KVMs and cloud environments such as AWS and DigitalOcean. - Delivered a final project deploying a secure WordPress site on Linux infrastructure.	
Artificial Intelligence Workshop <i>PDC NUST-PNEC</i>	Jan 2023
- Participated in an applied workshop introducing AI fundamentals and practical implementations. - Used Python libraries for text classification and image recognition tasks.	
Computer Vision Workshop <i>PDC NUST-PNEC</i>	Mar 2023
- Completed training on Python-based Computer Vision using Convolutional Neural Networks (CNNs). - Applied image detection and classification techniques using AI models.	
Basic Management Course <i>PNS RAHNUMA</i>	Aug 2014
- Developed strong administration, leadership, and presentation skills. - Enhanced public speaking and communication capabilities. - Learned principles of efficient office administration and management.	
Logistics, Management, and Naval Law <i>PNS KARSAZ</i>	Jun 2015
- Trained in naval logistics and supply chain management. - Studied naval law, financial compliance, audit, and regulatory frameworks.	
CBRNDC & Fire Fighting Course <i>PNS KARSAZ</i>	Jul 2015
- Gained understanding of CBRN warfare principles and fire safety operations. - Trained on firefighting equipment, organizational structure, and damage control drills.	
Junior Officer Staff Duties Course <i>PNS RAHNUMA</i>	Mar 2016
- Enhanced research, presentation, and correspondence skills. - Developed deep understanding of command, control, and staff duties in a military organization.	
Naval Operations Course <i>PNS BAHADUR</i>	Jan – Sep 2016
- Trained in secure communications, physical security, electronic warfare, and information warfare. - Gained proficiency in surface, above-water, and underwater operations. - Completed modules in maritime and astro-navigation, small arms handling, and ship formations.	
Helicopter Controller Course <i>PNS MEHRAN</i>	2017
- Trained in secure air traffic control for military aviation operations. - Applied secure communication protocols for air-sea coordination. - Learned operational procedures for air asset tasking and command control at sea.	
Bridge Watch Keeping Certificate <i>PNS TARIQ (Type 21)</i>	2017
- Qualified in ship handling, navigation, and formation management. - Trained in radio and visual communications, damage control, and electronic warfare. - Gained experience in crew leadership and shipboard HR management.	
Basic Navy SEAL Course <i>NSOTC</i>	2018
- Completed all modules of the course except the final phase. - Trained in physical security, VIP protection, secure communication, and close-quarter combat. - Conducted anti-terrorist, boarding, and combat diving operations.	
Maritime Law Enforcement and Policing <i>HQ PMSA</i>	2019
- Completed training in maritime law enforcement, port and fishing control, and coastal security. - Studied anti-human trafficking and anti-drug operations within maritime jurisdictions.	

Language Proficiency

IELTS Academic

25 October 2024

British Council, IDP, Cambridge English

Overall Band 8.0

- Achieved an Overall Band Score of 8.0.
- Individual Band Scores: Listening 8.5 (C2), Reading 7.5 (C1), Writing 8.0 (C1), Speaking 7.5 (C1).
- Test Report Form Number: 24PK501964KHAM363A.

Russian Language Course (A1) – 72 Hours

28 October 2024

Ural State Pedagogical University - Centre for Open Education

Yekaterinburg, Russia

- Completed the course on the Russian language and culture (A1 level).
- Program: "Culture of Russia".
- Course duration: 02 September 2024 to 28 October 2024.
- Total volume: 72 hours.

Spanish Language Course (A1)

03-28 Nov 2025

Centro de Lenguas Modernas (CLM), Universidad de Granada

Granada, Spain

- Completed the A1 level course for Spanish language and culture.
- Course duration: 03 November 2025 to 28 November 2025 i.e. 80 hours.

Miscellaneous Courses & Certifications

Life Book Mastery | *Mindvalley*

2020

- Developed skills in **life planning, time management, and career goal setting**.

Silva Ultra Mind Instructor | *Avlis Pro*

2019

- Certified as a **meditation instructor** specializing in stress reduction and mental resilience.

Certified Hypnotherapist | *First Institute of Dynamic Learning by Kamran Sultan*

2020

- Certified in **hypnotherapy techniques** for stress management and cognitive enhancement.
- Applied principles of **neuro-linguistic programming (NLP)** to improve communication and influence.

Fire & Glass Walk Course | *First Institute of Dynamic Learning by Kamran Sultan*

2020

- Trained participants in **mental resilience and overcoming physical limitations**.
- Successfully completed final assessment involving a **glass walk challenge**.

Projects

CodeGuardian | *Python, Docker, Ollama, Gemini CLI*

2024 – Present

- Integrated **Ollama** to enable **offline code review** workflows using open-source LLMs for enhanced privacy and local security.
- Implemented **multi-model support**, including Google **Gemini CLI** and OpenAI **Codex**, for comparative evaluation of code-analysis capabilities.
- Designed a modular pipeline enabling scalable AI-based static review and developer-assist features.

Ascon Encryption – C Implementation | *C, Cryptography*

2024

- Conducted **research and documentation** of **Ascon**, a NIST SP 800-232 Standardized lightweight authenticated encryption algorithm.
- Analyzed performance and correctness aspects of core primitives on **resource-constrained** systems.

Ascon Encryption – Rust Implementation | *Rust, Cryptography*

2024

- Performed **cryptographic research** and documentation for a Rust-based implementation prioritizing **memory-safety** and **performance**.
- Supported alignment with modern **lightweight AEAD standards** for embedded and IoT applications.

Offline $\LaTeX$ Editor & Customized Book Template (PN) | $\LaTeX$

Feb 2023 – Jul 2023

- Built a **fully offline, locally hosted $\LaTeX$** editor for secure content development.
- Developed **standardized $\LaTeX$ templates** for books and publications from scratch.
- Implemented offline open-source **book management** with **Calibre** and **Kavita Reader**.

Speech-to-Text for Urdu Language | *Bash, Python, Perl, C++, JSON, SQL*

Oct 2021 – Dec 2022

- Trained a **Hybrid HMM-DNN STT system** from scratch on a customized dataset at CPLC as part of MS CyS thesis.

- Automated **Linux-based pipelines** with services, cron jobs, and **Bash** scripts for data cleaning and model execution.
- Leveraged **Kaldi**, **DeepSpeech**, and **Vosk** frameworks for modeling and evaluation.

IoT-Based Temperature and Humidity Detector | *Bash, Python, MongoDB*

Feb – Jun 2021

- Completed an **end-to-end secure IoT project** for MS CyS, covering device, network, and application layers.
- Integrated **MQTT** with **TLS/SSL** in a **pub/sub** model using **Mosquitto** as the broker.
- Implemented data collection in **Prometheus (time-series)** and **MongoDB (NoSQL)**.
- Developed dashboards in **Grafana** for real-time visualization and monitoring.
- Performed **packet capture and security analysis** to validate sensor communication integrity.
- Programmed **ESP32** microcontroller with temperature/humidity sensors via **Arduino** on Linux and Windows.
- Containerized the PoC using **Docker** for reproducible deployment.

Recruitment Analysis System | *Python, HTML, CSS, Power BI*

2017–2018

- Delivered a **final-year MIS capstone** focused on recruitment analytics and decision support.
- Built a **proof of concept** for data-driven **recruitment analysis** and candidate evaluation.

Public Speaking

Pakistan's Wind Power Revolution: Turning Coastal Breezes into Energy | *News 99* 14 October 2025

- Featured on *Frontline with Tazmeen Javaid* discussing the potential of renewable energy in Pakistan and the role of the IntelliWind EU project on improving the Wind Energy sector, followed by a detailed article on medium.com.

Brain Drain and Bureaucracy in Pakistan | *News 99* 01 Aug 2025

- Featured on *Frontline with Tazmeen Javaid* discussing **brain drain**, bureaucracy, and the potential of **AI and blockchain** in public-sector reform.

Will Artificial Intelligence Take Jobs Or Create New Opportunities? | *News 99* 16 Apr 2025

- Appeared on *Frontline with Tazmeen Javaid* with Saim Safder, covering **AI's impact** on creativity, research, and the **future of work**.

The Impact of Pakistan's Slow Internet on the Economy | *News 99* 18 Feb 2025

- Discussed how **limited internet connectivity** affects productivity, competitiveness, and **economic growth**, highlighting digital infrastructure challenges and solutions.

DeepSeek and Impact on Other LLMs | *News 99* 30 Jan 2025

- Explored **DeepSeek** and strategic implications for **AI competitiveness**, including China-US dynamics in advanced AI.

State of Linux Kernel | *Cloud Native FM (Saim Safder)* 18 Jan 2025

- Joined **Cloud Native FM** podcast to discuss **Linux kernel** trends and **AI development** considerations.

Cyber Security for Kids | *Pakistan Navy* 20 Feb 2024

- Delivered an online session on **cybersecurity awareness** tailored for children and parents.

AI and Cyber Security | *AI Center of Pakistan* 20 Feb 2024

- Presented an applied session on **AI and cybersecurity**, focusing on risks, controls, and safe adoption.

Spendless Academy | *Remote Trainer* Aug 2018 – May 2022

- Launched online instruction prior to widespread remote learning, building early **distance-teaching** experience.
- Taught **speed reading**, **memory improvement**, **meditation**, and the **Silva UltraMind ESP** system.

Bahria University | *Workshop Facilitator* 2022

- Facilitated **Navigating the Digital World** workshop for student psychologists (Bachelor's).
- Strengthened **research skills**, **fact-checking**, and **information verification** practices.

Leadership / Extracurricular / Volunteer

Cloud Native Computing Foundation Islamabad Feb 2025 – Present

Developer Relations & Community Builder

Part-Time, Remote

- Contributed to **podcasts and content** (Cloud Native FM) on **Linux**, **AI**, **DevOps**, and **Cybersecurity**.
- Coordinated and **hosted community events** focused on the **CNCF stack**.
- Designed and **delivered training** on core **CNCF technologies**.

Model United Nations in NUST PNEC (MUNINP) 2015–2016

Director General

NUST-PNEC

- Organized NUST PNEC's **first nationwide MUN** as Director General.
- Managed a conference with **600+ delegates**, including international participants.
- Delivered two major events: **Grand Carnival** and **Qawali Night**.
- Participated as a delegate in **five MUNs** and received awards.
- Chaired more than **15 MUN committees** across Pakistan.

Hobbies

- | | | | |
|---------------------------|---------------------|----------------|-------------------|
| • Music Composition | • Research Work | • Basketball | • Power Lifting |
| • Piano | • Course Designing | • Squash | • Coaching |
| • Guitar | • Chess | • Table Tennis | • Public Speaking |
| • Singing | • Swimming & Diving | • Badminton | • YouTube |
| • Blogging | • Sailing | • Cricket | • Podcasts |
| • Article & Novel Writing | • Football | • Calisthenics | • Mentoring |

Awards

Tamgha-e-Marka-e-Haq | *Pakistan Navy*

16 May 2025

- Awarded the **Tamgha-e-Marka-e-Haq** medal in recognition of service and operational readiness during the **Pakistan–India standoff** (Marka-e-Haq Day, 10 May 2025). (*Auth: YGPP3KA dated 16 May 2025*)

Pakistan Diamond Jubilee Medal | *Pakistan Navy*

26 Jan 2023

- Received the **Pakistan Diamond Jubilee 75th Independence Day Medal** for distinguished service and dedication to duty.

10 Years Long Service Medal | *Pakistan Navy*

01 Jan 2022

- Honored with the **10 Years Long Service Medal** for a decade of continuous and exemplary service in the Pakistan Navy.

Tamgha-e-Azm | *Pakistan Navy*

23 Mar 2018

- Conferred the **Tamgha-e-Azm Military Service Medal**, recognizing **exceptional bravery, commitment, and service** in operations against militancy and terrorism.
- The medal, established by the **Ministry of Defence (2018)**, honors members of the Armed Forces, Civil Armed Forces, and Law Enforcement Agencies serving on active strength between **Dec 11, 2001 and Mar 23, 2018**.

Publications

Warrior | Philosopher: Rise of Reyes

Novel

September 2025

Amazon Kindle Direct Publishing

Abstract: A contemporary sports drama that fuses mixed martial arts realism with philosophical reflection. The novel follows Tony “The Spartan” Reyes, an MMA contender who rebuilds his career and identity after personal loss, while confronting rivalry, grief, and self-mastery. Through disciplined narrative building and thematic exploration of resilience and stoic thought, the work blends high-intensity fight culture with questions of meaning, identity, and human growth. ASIN: B0FT6DRV3D. ISBN: 978-84-09-77559-0.

Open For Business: Harnessing the Power of Open Source in Corporate World

Book

October 2025

Taylor & Francis

Abstract: Open for Business by Sage Khan & Rahim Ali explores the transformative role of open-source software in the corporate landscape. It examines the evolution of software licensing, from the early days of copyright and proprietary models, championed by figures like Bill Gates, to the rise of open-source principles driven by the Free Software Foundation and the GNU Project. The book explores how open-source projects, supported by the Linux Foundation, have become integral to modern IT infrastructure, promoting innovation, security, and cost efficiency. It highlights the growing specialization in fields like DevOps, AI, and cybersecurity, and discusses the challenges companies face in maintaining in-house expertise, often leading to outsourcing for convenience. Khan critically assesses the impact of corporate intervention on open-source projects, using case studies of Red Hat, HashiCorp, and others to illustrate how business decisions can sometimes undermine the open-source ethos. Despite these challenges, the book highlights the resilience of the open-source community through initiatives such as project forks and alternative licensing models. “Open for Business” advocates for the adoption of open-source solutions to enhance organizational learning and flexibility while balancing corporate needs for scalability and support. It calls for a renewed commitment to the principles of transparency, collaboration, and freedom in the software industry. DOI: 10.1201/9781003536314. ISBN: 9781003536314.

Design and Development of Next-Generation CNC Robotic Machine

Research Paper

Apr 2025

SES Journal, Zenodo DOI

Abstract: The rapid evolution of manufacturing technologies has driven the need for more efficient, precise, and adaptive systems. This paper “Design and Development of Next-Generation CNC Robotic Machine Integrated with Laser Technology for Smart Manufacturing” presents the design and development of a next-generation Computer Numerical Control (CNC) robotic machine integrated with laser technology for smart manufacturing. By combining the advanced capabilities of CNC robotics with the precision of laser systems, the proposed system enhances manufacturing flexibility, accuracy, and automation across diverse industries. State-of-the-art robotic arms, high-power laser tools, and intelligent control systems are integrated to enable multi-material processing, rapid prototyping, and complex part production. The system is engineered with a focus on energy efficiency, safety, and adaptability, incorporating real-time data analytics and predictive maintenance features that align with Industry 4.0 standards. Despite challenges regarding integration complexity and cost, experimental validations demonstrate that this innovative CNC robotic machine offers significant promise in transforming manufacturing practices into smarter, more agile production environments. Future work is anticipated to further optimize the system through the integration of advanced AI and machine learning techniques for adaptive process control.

IOT-based Smart Building Management

Research Paper

21 Apr 2025

SES Journal, Zenodo DOI

Abstract: A Dual Approach to Automation and Safety with Real-Time Monitoring and Alerts - The integration of Internet of Things (IoT) technologies into building management systems has revolutionized the development of smarter, more efficient and safer environments. This paper presents a dual approach to smart building management, emphasizing both automation and safety through real-time monitoring and alert mechanisms. The automation system utilizes bidirectional ultrasonic sensors to monitor building occupancy and, in conjunction with temperature sensors, dynamically adjusts appliances such as lights, fans, air conditioners, and heaters to optimize comfort and energy efficiency. A notable feature of the system is its ability to enable remote monitoring and control of appliances via SMS communication. On the safety front, the system incorporates a suite of IoT sensors including smoke, flame, gas, temperature, and ultrasonic sensors to detect hazards such as fires or gas leaks. In response to such events, the system automatically disconnects

circuit breakers, activates alarms for evacuation, and dispatches real-time alert messages to occupants' mobile devices. This dual-function system enhances operational efficiency, reduces energy consumption, and significantly improves building safety. The paper details the design, implementation, and practical benefits of this integrated IoT-based smart building management solution. ISSN (e) 3007-3138 (p) 3007-312X

Leveraging AI in Pharmacy & Clinical Pharmacy Transformative

07 May 2025

Review Paper

SES Journal, DOI

Abstract: Leveraging Artificial Intelligence in Pharmacy and Clinical Pharmacy Transformative Innovations for Precision, Operational Efficiency, and Enhanced Patient-Centered Care covers that By improving accuracy, operational effectiveness, and patient-centered care, the use of artificial intelligence (AI) in clinical pharmacy and pharmacy is transforming healthcare. Drug discovery, customized medicine, and pharmaceutical treatment management are being optimized by AI-driven advances, including machine learning algorithms, natural language processing, and predictive analytics. AI-powered decision support systems in clinical pharmacies increase workflow efficiency, decrease adverse medication reactions, and improve prescription accuracy. Real-time patient monitoring, AI-assisted drug use evaluations, and automated dispensing robots all help to optimize resources and enhance therapeutic results. Additionally, AI improves drug adherence through digital health interventions such as chatbots, virtual assistants, and smartphone apps that offer instructional help and customized reminders. Drug safety surveillance is strengthened by the early detection of adverse drug responses made possible by the incorporation of AI in pharmacovigilance. Despite these developments, competent AI implementation requires addressing issues including data protection, legal compliance, and ethical considerations. This paper examines how artificial intelligence (AI) is changing pharmacy and clinical pharmacy, highlighting significant advancements, advantages, and difficulties, as well as how it might change the way healthcare is delivered. Pharmacists may adopt a more proactive, data-driven strategy by utilizing AI, which will eventually enhance patient safety, treatment results, and healthcare efficiency. To enable AI's responsible and successful incorporation into pharmacy practice, future research should concentrate on improving AI algorithms, encouraging multidisciplinary cooperation, and creating strong regulatory frameworks. To satisfy the changing needs of precision medicine and patient-centric healthcare, this study emphasizes the critical necessity for ongoing developments in AI-driven pharmacy solutions. ISSN Print: 2960-2793, ISSN Online: 2960-2807

Code-Switched Urdu ASR for Noisy Telephonic Environment

Jul 2023

Research Paper

Arxiv, S2IS

Abstract: Call Centers have huge amount of audio data, which can be used for achieving valuable business insights and transcription of phone calls is a tedious task when done manually. An effective Automated Speech Recognition system can accurately transcribe these calls for easy search through call history for specific context and content, allowing automatic call monitoring, improving QoS through keyword search and sentiment analysis. ASR for Call Center requires more robustness as the telephonic environment are generally noisy. Moreover, there are many low-resourced languages that are on verge of extinction, which can be preserved with the help of Automatic Speech Recognition Technology. Urdu is the 10th most widely spoken language in the world, with 231,295,440 worldwide, still remains a resource constrained language in ASR. Regional call-center conversations operate in the local language, with a mix of English numbers and technical terms generally causing *code-switching* problem. Hence, this paper describes an implementation framework of a resource-efficient Automatic Speech Recognition/ Speech to Text System in a noisy call-center environment using Chain Hybrid HMM and CNN-TDNN for Code-Switched Urdu Language. Using Hybrid HMM-DNN approach allowed us to utilize the advantages of Neural Network with less labelled data. Adding CNN with TDNN has been shown to work better in a noisy environment due to CNN's additional frequency dimension, which captures extra information from noisy speech, thus improving accuracy. We collected data from various open sources and labelled some of the unlabelled data after analysing its general context and content from Urdu language as well as from commonly used words from other languages, primarily English and were able to achieve WER of 5.2% with noisy as well as clean environment in isolated words or numbers as well as in continuous spontaneous speech.

Urdu ASR for Mal-intended Speech

Oct 2022

Thesis NUST-PNEC

Research Gate

- Thesis work for Master's Degree
- Developed latex proficiency

- Practical implementation done in CPLC call center
- Achieved 95% Speech to Text Transcription accuracy on Noisy Telephonic Urdu custom CPLC data using Hybrid HMM-CNN-TDNN Model with Lattice Free MMI Objective Function

- **Understanding Recursive Nature of Fractals** — Explored the mathematical foundations of self-similarity and recursive algorithms, highlighting their applications in procedural generation and computer graphics.
- **Fractals in Nature** — Analyzed biological and geological patterns through the lens of fractal geometry, demonstrating how simple recursive rules manifest in complex natural systems like river networks and lung structures.
- **Charter of Economy and Technology** — Proposed a non-partisan framework for Pakistan's IT sector, emphasizing the need for long-term policy consistency to decouple technological growth from political volatility.
- **Modernizing Legacy Apps with Docker & Kubernetes** — Outlined architectural strategies for transitioning monolithic systems to cloud-native microservices, focusing on container orchestration and high-availability deployment.
- **From the First Hard Fall to the Echo of Legend** — A philosophical exploration of leadership and resilience, drawing parallels between historical excellence and the requirements for modern technical and ethical leadership.
- **AI in Project Management Practices** — Discussed the integration of generative AI into Agile workflows, specifically focusing on automated sprint planning, resource allocation, and predictive risk assessment.
- **AI Vibe: Coding and Policy Enforcement** — Examined the 'Vibe Coding' phenomenon, balancing the speed of natural language development with the necessity of automated security guardrails and architectural linting.
- **Open Source and AI Synergy** — Argued for the necessity of open-source ecosystems in AI development to ensure transparency, ethical standards, and the democratization of high-performance models.
- **The DevOps of AI Ecosystem** — Defined the MLOps lifecycle, covering the bridge between model training and production-grade deployment through automated evaluation and continuous monitoring.
- **Standard Documentation for Pakistani Citizens** — Compiled a comprehensive logistical guide for students and professionals, detailing the bureaucratic and technical requirements for international academic and professional mobility.
- **Braindrain & Bureaucracy** — Investigated the systemic causes of talent migration in Pakistan and advocated for digital-first policies and remote-work hubs to retain local intellectual capital.
- **Model Context Protocol: Hype, Anti-Pattern, or Standard?** — Analyzed the emerging open standard for LLM-data integration and its role in decoupling agentic workflows from proprietary interfaces.
- **Beyond ChatGPT with a UI: Why AI Wrappers Matter** — Investigated strategies for building defensible AI-native startups by leveraging vertical domain depth, proprietary data moats, and workflow integration.
- **Potential of Pakistan in the Renewable Sector** — Explored Pakistan's 2.9M MW solar potential and proposed policy reforms like smart-grid integration and localized manufacturing for energy independence.
- **Rulefiles in AI-Powered Development** — Outlined a paradigm shift in AI-assisted coding using version-controlled instruction sets to enforce architectural standards and security policies.
- **Ascon: The Tiny Titan of IoT Security** — Conducted a deep technical dive into the NIST-standardized lightweight cryptographic algorithm, focusing on efficiency in resource-constrained IoT devices.
- **Open Source in the Corporate World** — Discussed achieving technological sovereignty and operational agility through the strategic adoption of open-source architectures and enterprise toolsets.
- **DevOps for Data: Apache Spark on Containers** — Detailed the integration of big data workloads into CI/CD pipelines via containerized Spark, data-quality gates, and multi-tenant resource management on Kubernetes.